Truth-Table Quotients Do Not Close One-Tape MCSP Magnification: Small Fibers, Exact Cosets, and a Defect-Inflation Dichotomy

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Abstract

McKay, Murray and Williams showed that an $N^{1.01}$ one-tape lower bound for MCSP at a small threshold $2^{\mu_* n}$ implies $P \neq NP$, while Cheraghchi, Hirahara, Myrisiotis and Yoshida proved nearly quadratic one-tape lower bounds only for thresholds $2^{\mu n}$ with $\mu > 1/2$. We ask whether a truth-table encoding can transfer the latter to the former, and prove counting and additive-combinatorial barriers that hold for arbitrary, even uncomputable, maps. First, Lupanov's construction and circuit enumeration give $\log_2 |\mathcal{L}_S| = \Theta(\min\{2^n, S \log S\})$ for the set $\mathcal{L}_S$ of size-$\leq S$ functions; hence any map sending $\mathcal{L}_{2^{\mu_H n}}$ into the size-$2^{\mu_1 n'}$ functions on $n' = \kappa n + O(1)$ variables with $\mu_H > \kappa\mu_1$ has a fiber containing $2^{\Theta(n2^{\mu_H n})}$ easy functions. This excludes injective and small-fiber encodings, dummy padding, tensor lifts, symmetry canonicalisation and low-kernel affine maps, even with free composition. Second, iterated Harper isoperimetry shows that every large YES fiber is XOR-dispersed and that $\log_2(|\mathcal{L}_S + \mathcal{L}_S|/|\mathcal{L}_S|) \geq ((1-\mu)/2 - o(1))S$ for $S = 2^{\mu n}$; therefore no exact coset-fiber, in particular no exact affine, quotient bridges the regimes. Third, for every remaining map, either the average target XOR-defect over fiber pairs or the average fiber inflation on $\mathcal{L}_S + \mathcal{L}_S$ is $2^{\Omega(S)}$, with a split shown to be optimal for cardinality arguments. Limits: the exact decision-bit map survives all counting arguments, so excluding predicate-specific quotients requires charging their one-tape computation; gaps $S_1 \geq 2S_0$ remain open; two composition statements are conditional on an explicit model. Nothing here proves $P \neq NP$ or a new lower bound; small cases are checked exhaustively in the supplement.

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Problem and contribution

McKay, Murray and Williams [1] showed (in the form stated by Cheraghchi, Hirahara, Myrisiotis and Yoshida (CHMY) [2, Theorem 1]) that for some small constant $\mu_* > 0$, an $N^{1.01}$ lower bound for MCSP$[2^{\mu_* n}]$ against one-tape machines with a one-way input tape implies $P \neq NP$. CHMY [2, Theorems 2 and 3] prove nearly quadratic lower bounds against the same model, but only for size thresholds $2^{\mu n}$ with $\mu > 1/2$. A natural attempt to close this gap is to *encode* truth tables: map an n-variable truth table to an n'-variable one so that the high-threshold YES instances become low-threshold YES instances, and pull the target algorithm back. This paper asks which encodings can possibly work, and shows that the counting and additive structure of the set of easy functions rules out every injective, small-fiber, symmetry, affine and coset-fiber encoding, and forces any remaining one to have exponentially large defect.

Unconditional results (proved here).

- *Theorem 1 (sharp semantic entropy and fiber-corrected barrier).* For $n^2 \leq S \leq 2^{n-2}$, $\log_2 |\mathcal{L}_S| = \Theta(\min\{2^n, S \log S\})$, and any map sending all size-$\leq S$ functions to size-$\leq T$ functions on n' variables has a fiber containing F easy source functions with $c \min\{2^n, S \log S\} - \log_2 F \leq C(T+1)\log_2(n'+T+2)$. With $n' = \kappa n + O(1)$, $S = 2^{\mu_H n}$, $T = 2^{\mu_1 n'}$ and $\mu_H > \kappa\mu_1$, some fiber has $2^{\Theta(n2^{\mu_H n})}$ easy functions. Corollaries 2–3: injective maps, maps with fibers of size $2^{o(n2^{\mu_H n})}$, canonicalisation by any $2^{\mathrm{poly}(n)}$ symmetry group, and affine maps with kernel dimension $o(n2^{\mu_H n})$ cannot bridge the CHMY regime to the MMW threshold, even if composition is free.

- *Proposition 5 (dispersed fibers).* Iterated Harper isoperimetry shows that every large source-YES fiber is XOR-dispersed: from any fixed member, all but a $2^{-\Omega(nT)}$ fraction of partners have XOR circuit complexity $S - o(S)$.
- *Proposition 6 (doubling of the easy set).* $\log_2(|\mathcal{L}_S + \mathcal{L}_S|/|\mathcal{L}_S|) \geq ((1-\mu)/2 - o(1))S$ for $S = \lfloor 2^{\mu n} \rfloor$. Consequently no exact coset-fiber quotient, in particular no exact affine quotient, bridges the fixed exponents.
- *Proposition 7 (defect-or-inflation dichotomy).* For every map whose YES image lies in the target easy set, either the average number of target XOR-images over fiber pairs or the average fiber inflation on $\mathcal{L}_S + \mathcal{L}_S$ is $2^{((1-\mu_H)/4-o(1))S}$; explicit maps show the half-exponent split cannot be improved by cardinality arguments alone, and that runtime alone does not control the geometry.
- *Supporting statements:* Proposition 4 (constant-factor gaps admit large stable junta fibers, excluded as YES-preserving quotients for sub-doubling gaps by Proposition 5); a tensor test forcing $\mu_H = t\mu_1$ for direct-product encodings that preserve both promise sides; and an exact accounting of dummy padding, whose best exponent $2\mu_1 < 1/2$ is far below 1.01.

Conditional results. Proposition 8 states, *assuming* that a materializer of exponent b and a target decider of exponent a compose in the CHMY model (an interface not supplied by [1] or [2]), that the route reaches the CHMY window exactly when $\max\{b, \kappa a\} < 2\mu_H$. Proposition 9 is an illustrative branch-forcing quotient for which every fresh-output materializer obeys $\max\{b, k\} \geq 2\mu_H$; it is specific to that quotient.

Not claimed. We do not prove $P \neq NP$, any new one-tape lower bound, or that every predicate-specific quotient fails: maps with non-coset, predicate-specific fibers (for example the exact decision-bit map) survive all counting arguments, and excluding them requires charging their computation. Gaps $S_1 \geq 2S_0$ remain open. The results are barriers for the encoding route, not lower bounds for MCSP.

Related work

Tools. Counting and Lupanov's upper bound [3; 4] are standard. Growth of the set of easy functions under point flips, via vertex isoperimetry in the hypercube [5], was used by Chow [6, Proposition 2] to compare the number of functions computable with $\gamma/2$ and γ gates; Proposition 5's growth estimate (30)–(31) is a quantitative version of that argument. Our contributions are its use for the sumset doubling bound (Proposition 6), the fiberwise dispersion of quotients (Proposition 5), and the defect–inflation factorization for arbitrary quotients (Proposition 7).

Magnification and its barriers. Magnification for MCSP was developed by Oliveira and Santhanam [7], Oliveira, Pich and Santhanam [8], McKay, Murray and Williams [1] (the one-tape/streaming threshold used here) and Chen, Jin and Williams [9] (all sparse NP languages). Chen, Hirahara, Oliveira, Pich, Rajgopal and Santhanam [10] identified the *locality barrier*: magnification theorems hold relative to small-fan-in oracles, while known lower-bound techniques extend to such oracle circuits. CHMY [2, Theorem 3] give a related oracle barrier specific to one-tape machines: no oracle algorithm with short queries can extend MMW-style magnification above $\mu = 1/2$. Our results are complementary and of a different type: they do not concern the magnification theorem or the lower-bound technique, but the *reduction* between thresholds, and they hold for arbitrary, even uncomputable, maps.

Threshold-changing reductions. Kabanets and Cai [11] studied "natural" reductions to MCSP, whose output size and threshold depend only on input length; our maps are of this kind but go from MCSP to MCSP. We are not aware of prior work bounding fibers or additive doubling of truth-table quotients between MCSP thresholds (search of Semantic Scholar/OpenAlex/arXiv/ECCC terms "MCSP threshold reduction", "hardness magnification barrier", "sumset circuit complexity", 2026-10-25).

Relation to P versus NP. The question studied here sits on one explicit route to $P \neq NP$. By [1] in the form (2) below, an $N^{1.01}$ one-tape lower bound for MCSP$[2^{\mu_* n}]$ already implies $P \neq NP$, and [2] proves nearly quadratic one-tape lower bounds at thresholds $2^{\mu n}$ with $\mu > 1/2$. A threshold-lowering encoding that met the composition requirements described in the section on conditional composition boundaries would therefore turn existing lower bounds into a separation of P and NP. Our theorems show that every injective, small-fiber, symmetry, affine or coset-fiber encoding fails for counting and additive reasons alone, and that any remaining encoding must pay an exponential defect or inflation. They thus locate exactly what an encoding would have to achieve on this route; they do not settle the route.

Relation to rStation papers. `r/p_vs_np` currently contains no published papers. Our papers on `r/computer_science` concern symmetric-of-threshold counting [12] and ordered (OBDD) width of exponential-threshold MCSP [13]; the latter is a restricted-model lower bound, and the present paper contains no OBDD results.

Setup

Let

$$L_{\mu,n} = \{f : \{0,1\}^n \to \{0,1\} : \mathrm{CC}(f) \le \lfloor 2^{\mu n} \rfloor\}, \qquad N = 2^n. \tag{1}$$

The fixed Cheraghchi–Hirahara–Myrisiotis–Yoshida–McKay–Murray–Williams endpoint supplies a non-numerically specified constant $\mu_* > 0$ such that

$$L_{\mu_*} \notin \mathrm{DTIME}_1[N^{1.01}] \quad \Longrightarrow \quad P \neq NP. \tag{2}$$

Here $\mathrm{DTIME}_1[t]$ is the CHMY model: a deterministic $O(1)$-state Turing machine with one one-way read-only input tape and one initially blank two-way read/write work tape, running in $O(t(N))$ steps on the canonical N-bit truth table.

The constant can be fixed directly from the published argument. Under $P = NP$, MMW [1, Theorem 1.3] and CHMY [2, Lemma 9 and the proof of Theorem 1] give a fixed polynomial q such that, for every admissible time-constructible s,

$$\mathrm{MCSP}[s] \in \mathrm{DTIME}_1[Nq(s)]. \tag{2a}$$

Choose $C, d \ge 1$ with $q(t) \le Ct^d$, and then fix

$$0 < \mu_* < \min\{\tfrac{1}{4}, 0.01/d\}. \tag{3}$$

For $s(n) = \lfloor 2^{\mu_* n} \rfloor$, equation (2a) runs in $O(N^{1+d\mu_*}) = O(N^{1.01})$. Hard-wiring the finitely many exceptional lengths and taking the contrapositive gives (2). Thus μ_* is fixed only existentially; no numerical value is claimed.

In the parameterized structural statements below, μ_1 denotes a free target-threshold exponent. Whenever the magnification endpoint (2) is invoked, it is specialized to $\mu_1 = \mu_*$.

The Cheraghchi–Hirahara–Myrisiotis–Yoshida lower-bound regime applies above threshold exponent $1/2$ [2, Theorem 3]. We test whether a truth-table encoding can bridge the two regimes.

Fiber-corrected threshold conservation

We use the CHMY circuit convention on both sides: bounded fan-in, unbounded fan-out, basis $\{\mathsf{AND}, \mathsf{OR}, \mathsf{NOT}\}$, and size equal to the number of gates, including NOT gates. In particular, output complementation can change the size by one and is not assumed to preserve an exact threshold. The next counting bound does not assume that the map is efficient, local, linear, or systematic.

Theorem 1 (sharp semantic entropy and fiber-corrected barrier)

Let $S = S(n)$ and $T = T(n)$, where $n^2 \le S \le 2^{n-2}$, and let

$$Q_n : \{0,1\}^{2^n} \longrightarrow \{0,1\}^{2^{n'}} \tag{3a}$$

be arbitrary. If

$$\mathrm{CC}(f) \le S \quad \Longrightarrow \quad \mathrm{CC}(Q_n(f)) \le T \qquad \text{for every } f : \{0,1\}^n \to \{0,1\}, \tag{3b}$$

and

$$F = \max_y |\{f : \mathrm{CC}(f) \le S,\ Q_n(f) = y\}|, \tag{3c}$$

then there are constants $c, C > 0$, depending only on the circuit basis, such that for all sufficiently large n,

$$c \min\{2^n, S\log_2 S\} \le \log_2 |\mathcal{L}_S| \le C \min\{2^n, S\log_2 S\}, \qquad \mathcal{L}_S = \{f : \mathrm{CC}(f) \le S\},$$

and

$$\boxed{c \min\{2^n, S\log_2 S\} - \log_2 F \le C(T+1)\log_2(n' + T + 2).} \tag{3d}$$

In particular, fix $0 < \mu_1, \mu_H < 1$ and $\kappa > 0$. If $n' = \kappa n + O(1)$, $S = \lfloor 2^{\mu_H n} \rfloor$, and $T = \lfloor 2^{\mu_1 n'} \rfloor$, then $\mu_H > \kappa\mu_1$ forces

$$\boxed{F = 2^{\Theta(n 2^{\mu_H n})}.} \tag{3e}$$

Proof

The Lupanov upper bound [3; 4, Chapter 1] gives a basis-dependent constant $A > 0$ such that every Boolean function on r variables has circuit complexity at most

$$A\frac{2^r}{r} \tag{3f}$$

for every sufficiently large r. A convention in which constants are not free changes this bound by only $O(1)$ gates.

Let $q = \lfloor \log_2 S \rfloor$. The assumptions on S give $q \le n-2$, $q \to \infty$, and $A2^q/q \le S$ for all sufficiently large n. Choose the largest $r \in \{q, \dots, n\}$ such that (3f) is at most S. If $r = n$, every n-variable function lies in $\mathcal{L}_S$. If $r < n$, maximality gives

$$A\frac{2^{r+1}}{r+1} > S, \qquad \text{hence} \qquad 2^r > \frac{S(r+1)}{2A} \ge \frac{S\log_2 S}{4A}.$$

In both cases,

$$2^r = \Omega\big(\min\{2^n, S\log_2 S\}\big).$$

All 2^{2^r} functions on the first r variables, extended constantly in the remaining variables, belong to $\mathcal{L}_S$. This proves the semantic-entropy lower bound.

By the standard enumeration [4, Chapter 1], a topologically ordered circuit with at most U gates on m inputs has, at each gate, a constant number of choices for its type and at most $(m+U+2)^2$ choices for its predecessors. Including the output and all smaller sizes, the number of functions of circuit complexity at most U is at most

$$2^{O((U+1)\log(m+U+2))}. \tag{3g}$$

Applying (3g) with $(m, U) = (n, S)$, and using $S \ge n^2$ together with the trivial bound $\log_2 |\mathcal{L}_S| \le 2^n$, proves the matching semantic-entropy upper bound. The image of $\mathcal{L}_S$ contains at least $|\mathcal{L}_S|/F$ distinct functions, all lying in the target class counted by (3g) with $(m, U) = (n', T)$. This proves (3d).

For the fixed-exponent parameters, $\min\{2^n, S\log S\} = \Theta(n2^{\mu_H n})$, whereas

$$(T+1)\log(n'+T+2) = O(n2^{\kappa\mu_1 n}) = o(n2^{\mu_H n}).$$

Thus (3d) gives the lower bound in (3e), and $F \le |\mathcal{L}_S| = 2^{O(n2^{\mu_H n})}$ gives the matching upper bound. □

The theorem uses only the YES direction of an MCSP promise reduction. For injective maps, $F = 1$, and (3d) recovers $\mu_H \le \kappa\mu_1$. More generally, the same conclusion holds whenever

$$\log_2 F = o(n2^{\mu_H n}). \tag{3h}$$

This includes every advice-free exact encoding and every quotient whose source-low fibers have size at most $2^{\mathrm{poly}(n)}$.

Corollary 2 (the one-tape window is empty)

Suppose a map with $n' = \kappa n + O(1)$ preserves the YES side from $L_{\mu_H,n}$ to $L_{\mu_1,n'}$, where $\mu_H > 1/2$, $0 < \mu_1 < 1/4$, and its low-complexity fibers satisfy (3h). Even if a target one-tape computation can be composed with the map at zero additional exponent cost, the CHMY lower bound cannot exclude a target running time $(2^{n'})^{1.01}$. For the magnification conclusion, specialize the free target exponent to $\mu_1 = \mu_*$.

Indeed, such a composition would run on the source table in time $2^{(1.01\kappa+o(1))n}$, while CHMY can exclude it only if

$$1.01\kappa < 2\mu_H. \tag{3i}$$

Theorem 1 instead gives

$$2\mu_H \le 2\kappa\mu_1 < \frac{\kappa}{2} < 1.01\kappa. \tag{3j}$$

Thus the required parameter interval is empty before any strict one-tape simulation overhead is charged. □

Corollary 3 (symmetry and affine quotients)

Canonicalization that identifies only functions in a common orbit of a transformation family $\mathcal{G}_n$ has $F \le |\mathcal{G}_n|$. Hence no symmetry family of size $2^{\mathrm{poly}(n)}$, including input permutations, input negations, affine changes of variables, and output complementation when allowed by the circuit convention, can bridge the threshold exponents.

If $Q_n(f) = A_n f + b_n$ is affine over $\mathbb{F}_2$ and $d_n = \dim \ker A_n$, then $F \le 2^{d_n}$. In the regime $\mu_H > \kappa\mu_1$, Theorem 1 therefore requires

$$\boxed{d_n = \Omega(n2^{\mu_H n}).} \tag{3k}$$

If the two sides of the source MCSP promise are constant on every fiber, and $\mathcal{L}_S = \{f : \mathrm{CC}(f) \le S\}$, then necessarily

$$\ker A_n \le \mathrm{Stab}(\mathcal{L}_S) := \{h : \mathcal{L}_S + h = \mathcal{L}_S\}. \tag{3l}$$

Indeed, the fibers are cosets of $\ker A_n$, so fiber-monochromaticity is precisely invariance of $\mathcal{L}_S$ under this kernel. Equations (3k)–(3l) are necessary, not sufficient. The zero fiber alone does not yield a contradiction. The Lupanov junta construction in Theorem 1, together with its matching entropy upper bound, gives

$$\max\{\dim K : K \le \mathbb{F}_2^{2^n},\ K \subseteq \mathcal{L}_S\} = \Theta(\min\{2^n, S\log S\}).$$

Because the zero function belongs to $\mathcal{L}_S$, its stabilizer is itself contained in $\mathcal{L}_S$. Thus an exact affine bridge in the fixed-exponent regime would require a stabilizer subspace of the largest possible order $\Theta(S\log S)$, while its quotient outputs would still have to remain below the target circuit threshold.

The phrase *in the bridging regime* is essential here. There is no uniform bound

$$\dim \mathrm{Stab}(\mathcal{L}_S) = o(\min\{2^n, S\log S\})$$

throughout the entire range of Theorem 1. Indeed, the Lupanov upper bound gives

$$\max_f \mathrm{CC}(f) \le (1 + o(1))\frac{2^n}{n} + O(n). \tag{18}$$

Thus, for all sufficiently large n, taking $S = 2^{n-2}$ makes $\mathcal{L}_S$ the whole truth-table space and

$$\dim \mathrm{Stab}(\mathcal{L}_S) = 2^n = \Theta(\min\{2^n, S\log S\}). \tag{19}$$

The open case relevant to (3k) is instead $S = \lfloor 2^{\mu n}\rfloor$ for a fixed $0 < \mu < 1$, far below this saturation threshold.

There is an exact spectral formulation of that open case. For $V = \mathbb{F}_2^{2^n}$, let $\widehat{1_{\mathcal{L}_S}}$ denote the unnormalized Fourier transform on V. Translation invariance gives

$$\mathrm{Stab}(\mathcal{L}_S) = \left(\mathrm{span}_{\mathbb{F}_2} \mathrm{Supp}\,\widehat{1_{\mathcal{L}_S}}\right)^{\perp}. \tag{20}$$

Consequently the desired bridging-regime estimate is equivalent to

$$\mathrm{rank}_{\mathbb{F}_2} \mathrm{Supp}\,\widehat{1_{\mathcal{L}_S}} = 2^n - o(\min\{2^n, S\log S\}). \tag{21}$$

Also, $\mathcal{L}_S$ is a union of stabilizer cosets, so

$$2^{\dim \mathrm{Stab}(\mathcal{L}_S)} \mid |\mathcal{L}_S|, \qquad \dim \mathrm{Stab}(\mathcal{L}_S) \le \nu_2(|\mathcal{L}_S|). \tag{22}$$

Ordinary circuit enumeration bounds $|\mathcal{L}_S|$, but not its 2-adic valuation or the rank in (21), and therefore does not determine the stabilizer itself. Proposition 6 below nevertheless rules out the exact affine bridge by a different invariant: multi-step additive expansion.

Proposition 4 (constant-factor gaps admit large stable fibers)

For thresholds $S_0 < S_1$, let $c_\oplus$ be the constant cost of XOR over the fixed basis (one may take $c_\oplus = 4$), and write

$$\mathcal{L}_{S_0} = \{f : \mathrm{CC}(f) \le S_0\}, \qquad \mathcal{H}_{S_1} = \{f : \mathrm{CC}(f) > S_1\}, \qquad B = \min\{S_0, S_1 - S_0 - c_\oplus\}. \tag{23}$$

If $n^2 \le B \le S_0 \le 2^{n-2}$, there is a linear $K \le V$ such that

$$K \subseteq \mathcal{L}_{S_0}, \qquad w(K) := \max_{k \in K} \mathrm{CC}(k) \le B, \qquad \dim K = \Theta(\min\{2^n, B \log B\}). \tag{24}$$

For this K,

$$\mathcal{L}_{S_0} + K \subseteq \mathcal{L}_{S_1}, \qquad (\mathcal{H}_{S_1} + K) \cap \mathcal{L}_{S_0} = \emptyset. \tag{25}$$

The dimension scale in (24) is optimal up to constants among all linear spaces with $w(K) \le B$. In particular, if $S_1 - S_0 \ge \rho S_0$ for a fixed $\rho > 0$, then

$$\dim K = \Theta_\rho(\min\{2^n, S_0 \log S_0\}). \tag{26}$$

Proof

Let K be the subspace of all functions depending only on the first r variables, with r chosen by the Lupanov construction in Theorem 1 using budget B. Then $K \subseteq \mathcal{L}_B \subseteq \mathcal{L}_{S_0}$, $w(K) \le B$, and $\dim K = 2^r = \Theta(\min\{2^n, B \log B\})$, proving (24).

For every f and k, XOR gives

$$\mathrm{CC}(f + k) \le \mathrm{CC}(f) + \mathrm{CC}(k) + c_\oplus. \tag{27}$$

By the definition of B, equations (24) and (27) give the first inclusion in (25). The second is equivalent to the first: if $h = f + k$ with $k \in K$, then also $f = h + k$.

$$2^{\dim K} \le |\mathcal{L}_B| \le 2^{O(\min\{2^n, B \log B\})} \tag{28}$$

for every linear K with $w(K) \le B$, by Theorem 1. This proves optimality. Under the fixed-factor gap, $B = \Theta_\rho(S_0)$, which gives (26). □

For the quotient by this K, the zero fiber contains exactly $2^{\dim K}$ low-complexity functions. Under a fixed-factor gap, its $\log_2 F = \Theta_\rho(\min\{2^n, S_0 \log S_0\})$ term is on the same asymptotic scale as the sharp source entropy in (3d), invalidating any little-o fiber hypothesis at the level of orders. This does not determine the leading constant and does not construct a quotient satisfying $Q(\mathcal{L}_{S_0}) \subseteq \mathcal{L}_T$, nor a one-tape implementation of such a map. Even if those interfaces were supplied, a solver for the promise separating $\mathcal{L}_{S_0}$ from $\mathcal{H}_{S_1}$ need not decide the gray region $S_0 < \mathrm{CC}(f) \le S_1$; the exact CHMY lower bound therefore does not rule it out.

Proposition 5 (Harper growth and almost-everywhere XOR-dispersed fibers)

Write

$$\Phi(n, t) = |\mathcal{L}_t|, \qquad N = 2^n, \qquad \ell_n = 2n + 3. \tag{29}$$

Fix $0 < \mu < 1$. There is a constant C_μ such that, for all sufficiently large n, uniformly for every integer $0 \le t \le 2^{\mu n}$,

$$\boxed{\log_2 \Phi(n, t + \ell_n) - \log_2 \Phi(n, t) \ge n - \log_2(t + 1) - C_\mu \ge (1 - \mu)n - C_\mu.} \tag{30}$$

Consequently, if $B \le S \le 2^{\mu n}$, then

$$\boxed{\log_2 \Phi(n, S) - \log_2 \Phi(n, B) \ge \left\lfloor \frac{S - B}{2n + 3} \right\rfloor ((1 - \mu)n - C_\mu).} \tag{31}$$

Now let Q be an arbitrary map from n-variable to n'-variable truth tables satisfying

$$Q(\mathcal{L}_S) \subseteq \mathcal{L}_{n',T}.$$

For every nonempty source-YES fiber, set

$$Y_y = Q^{-1}(y) \cap \mathcal{L}_S, \qquad D(y) = \max_{f,g \in Y_y} \mathrm{CC}(f \oplus g). \tag{32}$$

Every fiber satisfying

$$|Y_y| \geq \frac{\Phi(n,S)}{\Phi(n',T)} \tag{33}$$

obeys

$$\boxed{\left\lfloor \frac{(S-D(y))_+}{2n+3} \right\rfloor \big((1-\mu)n - C_\mu\big) \leq \log_2 \Phi(n',T) \leq C(T+1)\log_2(n'+T+2).} \tag{34}$$

In particular, every maximum source-YES fiber obeys (34), and

$$\boxed{D(y) \geq S - O_\mu\big(n + (T+1)\log_2(n'+T+2)\big).} \tag{35}$$

Put

$$U = \log_2 \Phi(n',T), \qquad a_\mu = (1-\mu)n - C_\mu.$$

For all sufficiently large n, every fiber satisfying (33) also obeys, for each fixed $f \in Y_y$ and every integer $0 \leq B \leq S$,

$$\boxed{\Pr_{g\sim \mathrm{Unif}(Y_y)}[\mathrm{CC}(f\oplus g) \leq B] \leq \min\left\{1, \frac{\Phi(n,B)\Phi(n',T)}{\Phi(n,S)}\right\} \leq 2^{-\left(a_\mu\left\lfloor \frac{S-B}{\ell_n}\right\rfloor - U\right)_+}.} \tag{35a}$$

Equivalently, for $\lambda \geq 0$, set

$$q_\lambda = \left\lceil \frac{U+\lambda}{a_\mu} \right\rceil, \qquad B_\lambda = S - \ell_n q_\lambda.$$

Whenever $B_\lambda \geq 0$,

$$\boxed{\Pr_{g\sim \mathrm{Unif}(Y_y)}[\mathrm{CC}(f\oplus g) \leq B_\lambda] \leq 2^{-\lambda}.} \tag{35b}$$

For the fixed-exponent bridging parameters of Theorem 1, this becomes

$$\boxed{D(y) \geq S - O_{\mu_H}(n2^{\kappa\mu_1 n}) = S - o(S).} \tag{36}$$

Moreover, $U = O(nT)$ and $a_{\mu_H} = \Theta(n)$. Taking $\lambda = cnT$ for any fixed $c > 0$ in (35b) gives, uniformly for every fixed $f \in Y_y$,

$$\Pr_g[\mathrm{CC}(f\oplus g) > B_\lambda] \geq 1 - 2^{-cnT}, \qquad B_\lambda = S - O_{\mu_H,c}(nT) = S - o(S). \tag{36a}$$

If $Q(f) = Af + b$ is affine with kernel K, then the same conclusion gives

$$\boxed{w(K) \geq S - O_{\mu_H}(n2^{\kappa\mu_1 n}).} \tag{37}$$

If, in addition, $\mathcal{L}_S$ is a union of fibers, as two-sided exact promise preservation requires, then

$$\boxed{\frac{|K\cap \mathcal{L}_B|}{|K|} \leq 2^{-\left(a_\mu\left\lfloor \frac{S-B}{\ell_n}\right\rfloor - U\right)_+} \qquad (0 \leq B \leq S).} \tag{37a}$$

Thus in the fixed-exponent regime all but a $2^{-\Omega(nT)}$ fraction of the kernel has circuit complexity at least $S - O_{\mu_H}(nT)$.

Proof The circuit count (3g) supplies a basis-dependent constant C_0 such that

$$\log_2 \Phi(n,t) \leq C_0(t+1)\log_2(n+t+2).$$

Set $D_\mu = 4C_0/(1-\mu)$ and $r = \lfloor D_\mu(t+1)\rfloor$. Since $t \leq 2^{\mu n}$, for all sufficiently large n,

$$\log_2 \binom{N}{r} \geq r\log_2 \frac{N}{r} \geq 2C_0 n(t+1) \geq \log_2 \Phi(n,t). \tag{38}$$

Let I be the simplicial-order initial segment of the N-cube with $|I| = \Phi(n,t)$. Harper's vertex-isoperimetric theorem [5] says that I minimizes the closed neighborhood, and hence the external vertex boundary, among sets of this size.

Equation (38) gives $|I| \le \binom{N}{r} \le |B_r|$, where $B_r = \{x \in \mathbb{F}_2^N : |x| \le r\}$, so $I \subseteq B_r$. The initial segment is a downset, including when its top layer is only partially filled. Therefore the number of upward edges from I to its complement is

$$e^{\uparrow}(I, \mathbb{F}_2^N \setminus I) = N|I| - 2\sum_{x\in I} |x| \ge (N-2r)|I|. \tag{39}$$

Every external-boundary vertex has weight at most $r+1$, and hence receives at most $r+1$ such edges. Thus

$$\frac{|\partial\mathcal{L}_t|}{\Phi(n,t)} \ge \frac{|\partial I|}{|I|} \ge \frac{N-2r}{r+1} \ge \frac{N}{2(D_\mu+2)(t+1)}. \tag{40}$$

Changing one truth-table bit at $a \in \{0,1\}^n$ means replacing f by $f \oplus \delta_a$. In the CHMY convention, δ_a uses at most n input negations and $n-1$ AND gates, while

$$f \oplus \delta_a = (f \vee \delta_a) \wedge \neg(f \wedge \delta_a)$$

uses four more gates. Hence $\mathcal{L}_t \cup \partial\mathcal{L}_t \subseteq \mathcal{L}_{t+2n+3}$. Taking logarithms in (40) proves (30). (This point-flip plus isoperimetry growth argument, including its iteration between two thresholds, is due to Chow [6, Proposition 2]; the new content of Propositions 5–7 is its application to fibers of quotients and to the sumset $\mathcal{L}_S + \mathcal{L}_S$.) Iterating at $B, B+\ell_n, \dots, B+(q-1)\ell_n$, where $q = \lfloor (S-B)/\ell_n \rfloor$, proves (31). The floor is essential when $S - B < 2n+3$.

Fix a fiber satisfying (33) and an $f_* \in Y_y$. The map

$$g \longmapsto g \oplus f_*$$

is injective from Y_y into $\mathcal{L}_{D(y)}$. Hence

$$\Phi(n, D(y)) \ge |Y_y| \ge \frac{\Phi(n,S)}{\Phi(n',T)}.$$

If $D(y) \le S$, apply (31) with $B = D(y)$; if $D(y) > S$, the positive-part formulation is immediate. This proves (34). A maximum source-YES fiber satisfies (33) because $|Q(\mathcal{L}_S)| \le \Phi(n',T)$. Rearranging the floor gives (35), and (36) follows from $n' = \kappa n + O(1)$, $T = 2^{\kappa\mu_1 n + O(1)}$, and $\mu_H > \kappa\mu_1$.

For the pointwise tail, fix $f \in Y_y$. The same injective translation bounds the number of partners at XOR complexity at most B:

$$|\{g \in Y_y : \mathrm{CC}(f \oplus g) \le B\}| \le \Phi(n,B).$$

Divide by (33), and then apply (31), to obtain (35a). The definitions of q_λ and B_λ make $\lfloor (S - B_\lambda)/\ell_n \rfloor = q_\lambda$ and $a_\mu q_\lambda - U \ge \lambda$, proving (35b). The fixed-exponent circuit count gives $U = O(nT)$, so (36a) follows.

Finally, suppose Q is affine. If $w(K) > S$, (37) is immediate. Otherwise $Q^{-1}(b) = K \subseteq \mathcal{L}_S$, every affine fiber has size at most $|K|$, and $D(b) = w(K)$. Thus $Y_b = K$ is a maximum source-YES fiber, and (37) follows from (36). If $\mathcal{L}_S$ is a union of fibers, then the zero fiber is wholly contained in $\mathcal{L}_S$, every nonempty source-YES fiber is a coset of size $|K|$, and $|K| \ge \Phi(n,S)/\Phi(n',T)$. Apply (35a) to $Y_b = K$ with $f = 0$ to obtain (37a) and its fixed-exponent consequence. □

Proposition 5 supplies the missing target-side test for the junta kernels in Proposition 4. Put $S_0 = S$, $r_n = S_1/S$, and

$$B_* = \min\{S, S_1 - S - c_\oplus\}.$$

Whenever Proposition 4 applies,

$$S - B_* = \max\{0, 2S + c_\oplus - S_1\}. \tag{41}$$

If the quotient by this K were also YES-preserving to the target threshold, then for fixed $1 < r < 2$, the identity $S - B_* = (2-r)S + O(1)$ would contradict both (35) and the stronger pointwise tail (35a) in the bridging regime. For $r = 2$ the residual is only $O(1)$, and for $r > 2$ it is zero for all sufficiently large n, so this argument gives no obstruction. More generally, if $r_n \uparrow 2$, the same junta kernel is excluded whenever

$$\boxed{2 - r_n = \omega\big(n 2^{-(\mu_H - \kappa\mu_1)n}\big).} \tag{42}$$

This closes the uniformly subthreshold-width junta escape for sub-doubling gaps; Proposition 4 by itself remains only a gap-stability statement, not a construction of a compressed YES-preserving image. Maximal-budget kernels at gaps at least two, and kernels with no bound below the source threshold, are not covered.

The stronger Harper growth still does not by itself control the exact stabilizer. To see the precise limitation, fix $S = \lfloor 2^{\mu n} \rfloor$, choose a sufficiently large constant C, set

$$r = \left\lfloor \log_2(S \log_2 S) \right\rfloor - C, \qquad H = \{g(x_1, \dots, x_r)\},$$

and let $A = H$. By Lupanov's bound and the choice of r, a sufficiently large fixed C ensures, for all sufficiently large n, that

$$w(H) \le S/2$$

and hence $H \subseteq \mathcal{L}_S$, while

$$\dim H = \Theta(S \log S), \qquad \mathrm{Stab}(A) = H.$$

Every nonzero function in H is constant on truth-table blocks of size $2^{n-r} \ge 4$. Thus no point indicator e_a, nor any $e_a + e_b$ with $a \ne b$, lies in H. The N boundary cosets $H + e_a$ are therefore pairwise disjoint and avoid A, so

$$|\partial A| = N|A|. \tag{43}$$

Moreover, the same point-flip circuit gives $A \cup \partial A \subseteq \mathcal{L}_{S+2n+3}$. Hence low-circuit containment, maximal stabilizer dimension, and even maximal standard-coordinate boundary growth are numerically compatible. This model is not itself a bridge candidate: its uniformly low width violates (35a) for any fixed-exponent YES-preserving quotient achieving the required compression. Its role is only to show that entropy and standard-coordinate boundary growth do not by themselves control the exact stabilizer. Distinguishing the actual set $\mathcal{L}_S$ from this model at the level of the stabilizer itself requires circuit-specific translation information, for example the zero set of

$$\tau_S(h) = |\{f \in \mathcal{L}_S : f + h \notin \mathcal{L}_S\}|,$$

or the Fourier-support and 2-adic interfaces in (20)–(22); further standard-coordinate isoperimetry cannot settle the exact stabilizer. The bridge question is stronger than the bare stabilizer question, because target compression also bounds the number of source-YES cosets. That extra condition is enough for the following conclusion.

Proposition 6 (Harper doubling rules out exact coset quotients)

Fix $0 < \mu < 1$, let $n^2 \le S \le 2^{\mu n}$, and put

$$A = \mathcal{L}_S, \qquad \mathbb{B}_q = \{h \in \mathbb{F}_2^N : |h| \le q\}, \qquad q = \left\lfloor \frac{S+4}{2n+3} \right\rfloor.$$

Choose a basis-dependent constant C_0 such that $\log_2 \Phi(n,t) \le C_0(t+1)\log_2(n+t+2)$, and set

$$D_\mu = \frac{4C_0}{1-\mu}, \qquad R = \lceil D_\mu(S+1) \rceil.$$

For all sufficiently large n,

$$\boxed{\log_2 \frac{|A+A|}{|A|} \ge q \log_2\!\left(1 + \frac{N - 2(R+q)}{R+q+1}\right) \ge q\big((1-\mu)n - C'_\mu\big).} \tag{44}$$

In particular, if $S = \lfloor 2^{\mu n} \rfloor$, then

$$\boxed{\log_2 \frac{|\mathcal{L}_S + \mathcal{L}_S|}{|\mathcal{L}_S|} \ge \left(\frac{1-\mu}{2} - o(1)\right) S.} \tag{45}$$

Suppose now that the fibers of Q are the cosets of a fixed linear subspace K, that A is a union of fibers, and that $Q(A) \subseteq \mathcal{L}_{n',T}$. Then necessarily

$$\boxed{q\big((1-\mu)n - C'_\mu\big) \le \log_2 |Q(A)| \le \log_2 \Phi(n', T).} \tag{46}$$

Consequently, no such exact coset quotient exists for the fixed-exponent bridging parameters of Theorem 1. In particular, no exact affine quotient exists in that regime. Independently of any target map, (44) also gives

$$\boxed{\dim \mathrm{Stab}(\mathcal{L}_S) \le \log_2 \Phi(n,S) - q\big((1-\mu)n - C'_\mu\big).} \tag{47}$$

Proof

Every truth table of Hamming weight $m \geq 1$ is the XOR of m point indicators. The first indicator costs at most $2n-1$ gates, and each subsequent indicator together with its XOR costs at most $2n+3$ more. Hence

$$\mathrm{CC}(h) \leq m(2n+3) - 4 \leq S \qquad (h \in \mathbb{B}_q), \tag{48}$$

with the zero table included trivially. Thus $\mathbb{B}_q \subseteq A$ and

$$A + \mathbb{B}_q \subseteq A + A. \tag{49}$$

Let I_0 be the simplicial-order initial segment with $|I_0| = |A|$. The circuit count and the choice of R, exactly as in (38), give $|I_0| \leq \binom{N}{R}$, so $I_0 \subseteq \mathbb{B}_R$. Write $I_j = I_0 + \mathbb{B}_j$. The upper shadow of the partial top layer of a simplicial initial segment is again an initial segment; hence every I_j is a downset contained in $\mathbb{B}_{R+j}$. Iterating Harper's one-step theorem gives

$$|A + \mathbb{B}_j| \geq |I_j|. \tag{50}$$

For $0 \leq j < q$, the upward-edge count for I_j is at least $(N - 2(R+j))|I_j|$. Each external-boundary vertex receives at most $R + j + 1$ such edges. Therefore

$$\frac{|I_{j+1}|}{|I_j|} \geq 1 + \frac{N - 2(R+j)}{R+j+1} \geq 1 + \frac{N - 2(R+q)}{R+q+1}. \tag{51}$$

Multiplying (51), then using (49)–(50), proves the first inequality in (44). Since $R + q = O_\mu(S)$, its logarithmic factor is at least $(1-\mu)n - C'_\mu$, uniformly in j. This proves the second inequality and (45).

For the quotient statement, write A as the union of $m = |Q(A)|$ distinct K-cosets. Then $A + A$ is a union of at most m^2 such cosets, so

$$\frac{|A+A|}{|A|} \leq m = |Q(A)|. \tag{52}$$

Equations (44) and (52) prove (46). Under the fixed-exponent parameters, $\log_2 \Phi(n', T) = O(nT) = o(S)$, whereas the left side of (46) is $((1-\mu_H)/2 - o(1))S$, a contradiction. Finally take $K = \mathrm{Stab}(A)$. Then A is a union of $|A|/|K|$ cosets, and the same argument rearranges to (47). □

Proposition 7 (XOR-defect or fiber-inflation dichotomy)

Let Q be an arbitrary map on the source truth-table space, not necessarily exact or efficient. Retain $A = \mathcal{L}_S$, and define

$$Y = Q(A), \qquad M = |Y|, \qquad A_y = A \cap Q^{-1}(y), \qquad \overline{F} = \frac{|A|}{M}.$$

Also put

$$M_+ = |Q(A+A)|, \qquad \overline{F}_+ = \frac{|A+A|}{M_+}, \qquad C_M = \binom{M+1}{2}.$$

Define

$$\begin{aligned}
\mathsf{X}_Q(A) &= \frac{M_+}{M}, \\
\mathsf{I}_Q^{\mathrm{av}}(A) &= \frac{\overline{F}_+}{\overline{F}}, \\
\mathsf{I}_Q(A) &= \frac{1}{\overline{F}} \max_w |(A+A) \cap Q^{-1}(w)|, \\
\mathsf{D}_Q^{\mathrm{av}}(A) &= \frac{1}{C_M} \left(\sum_{y \in Y} |Q(A_y + A_y)| + \sum_{\{y,z\} \in \binom{Y}{2}} |Q(A_y + A_z)| \right), \\
\mathsf{D}_Q(A) &= \max_{y,z \in Y} |Q(A_y + A_z)|.
\end{aligned} \tag{53}$$

Since $0 \in A$, one has $A \subseteq A + A$, and therefore $\mathsf{X}_Q(A), \mathsf{I}_Q(A), \mathsf{D}_Q(A) \geq 1$. The average occupancy factor $\mathsf{I}_Q^{\mathrm{av}}(A)$ is positive but can be below one, while

$$\mathsf{I}_Q^{\mathrm{av}}(A) \leq \mathsf{I}_Q(A), \qquad \mathsf{D}_Q^{\mathrm{av}}(A) \leq \mathsf{D}_Q(A).$$

Then

$$\boxed{\mathsf{X}_Q(A)\mathsf{I}_Q^{\mathrm{av}}(A) = \frac{|A+A|}{|A|}, \qquad \mathsf{X}_Q(A)\mathsf{I}_Q(A) \geq \frac{|A+A|}{|A|} \geq 2^{q((1-\mu)n - C'_\mu)}.} \tag{54}$$

Moreover,

$$\boxed{\frac{|A+A|}{|A|} \leq \mathsf{I}_Q^{\mathrm{av}}(A)\left(\frac{\mathsf{D}_Q^{\mathrm{av}}(A)(M+1)}{2} - 1 + \frac{1}{M}\right) \leq \mathsf{I}_Q(A)\left(\frac{\mathsf{D}_Q(A)(M+1)}{2} - 1 + \frac{1}{M}\right).} \tag{55}$$

If $Q(A) \subseteq \mathcal{L}_{n',T}$, then

$$\boxed{\log_2 \mathsf{D}_Q(A) + \log_2 \mathsf{I}_Q(A) \geq \log_2 \mathsf{D}_Q^{\mathrm{av}}(A) + \log_2 \mathsf{I}_Q^{\mathrm{av}}(A) \geq q((1-\mu)n - C'_\mu) - \log_2 \frac{M+1}{2} \geq q((1-\mu)n - C'_\mu) - \log_2 \Phi(n', T).} \tag{56}$$

For the fixed-exponent bridging parameters,

$$\boxed{\log_2 \mathsf{D}_Q(A) + \log_2 \mathsf{I}_Q(A) \geq \log_2 \mathsf{D}_Q^{\mathrm{av}}(A) + \log_2 \mathsf{I}_Q^{\mathrm{av}}(A) \geq \left(\frac{1-\mu_H}{2} - o(1)\right) S.} \tag{57}$$

Thus every surviving quotient has either $\mathsf{D}_Q(A) = 2^{\Omega(S)}$ or $\mathsf{I}_Q(A) = 2^{\Omega(S)}$. More precisely,

$$\boxed{\max\{\mathsf{D}_Q^{\mathrm{av}}(A), \mathsf{I}_Q^{\mathrm{av}}(A)\} \geq 2^{((1-\mu_H)/4 - o(1))S}, \qquad \max\{\mathsf{D}_Q(A), \mathsf{I}_Q(A)\} \geq 2^{((1-\mu_H)/4 - o(1))S}.} \tag{57a}$$

Proof

The occupied fibers of Q partition $A + A$, giving the exact factorization

$$\frac{|A+A|}{|A|} = \frac{M_+ \overline{F}_+}{M\overline{F}} = \mathsf{X}_Q(A)\mathsf{I}_Q^{\mathrm{av}}(A) \leq \mathsf{X}_Q(A)\mathsf{I}_Q(A). \tag{58}$$

This proves the first part of (54); its lower bound is (44).

For the localized defect, commutativity gives

$$Q(A+A) = \bigcup_{y \in Y} Q(A_y + A_y) \cup \bigcup_{\{y,z\} \in \binom{Y}{2}} Q(A_y + A_z).$$

Every one of the M diagonal image sets contains $Q(0)$. Counting that common value once, the remaining diagonal values, and then the unordered off-diagonal pairs gives

$$M_+ \leq 1 + \sum_{y \in Y} \big(|Q(A_y + A_y)| - 1\big) + \sum_{\{y,z\} \in \binom{Y}{2}} |Q(A_y + A_z)| = C_M \mathsf{D}_Q^{\mathrm{av}}(A) - M + 1. \tag{59}$$

Combine (58)–(59), divide by M, and use the two average-to-maximum inequalities following (53) to prove (55). Using $-1 + 1/M \leq 0$ in (55), taking logarithms, and applying $(M+1)/2 \leq M \leq \Phi(n', T)$ proves (56). Finally $\log_2 \Phi(n', T) = O(nT) = o(S)$, so (57) follows from (45). Exponentiating (57), the maximum of the two positive average factors is at least the square root of their product; this proves (57a). □

The two factors cannot be separated by cardinality alone. For an exact affine quotient, all four average and maximum factors equal one, recovering Proposition 6. At the opposite extreme, the constant map has $M = M_+ = 1$, $\mathsf{D}_Q^{\mathrm{av}}(A) = \mathsf{D}_Q(A) = 1$, and $\mathsf{I}_Q^{\mathrm{av}}(A) = \mathsf{I}_Q(A) = |A+A|/|A|$, so it saturates (55) but preserves only the YES side. The exact predicate map that sends A to one label and its complement to another has $M = 1$; in the regime here, (44) gives $M_+ = 2$, $\mathsf{D}_Q^{\mathrm{av}}(A) = \mathsf{D}_Q(A) = 2$, and $\mathsf{I}_Q^{\mathrm{av}}(A) = |A+A|/(2|A|)$, so the average bound in (55) is exact. It preserves both sides

combinatorially, but computing it is exactly the source decision problem. Therefore pure counting cannot eliminate arbitrary predicate-specific fibers; a further bridge theorem must constrain XOR defect or fiber inflation using the actual one-tape computation.

The exponent split in (57a) is also sharp for cardinality arguments. Partition $(A+A)\setminus A$ into L nearly equal nonempty blocks, map all of A to one label, and assign one further label to each block. Then $M=1, M_+=\mathsf{D}^{\mathrm{av}}_Q(A)=\mathsf{D}_Q(A)=L+1$, and, writing $R=|A+A|/|A|$,

$$\mathsf{I}^{\mathrm{av}}_Q(A)=\frac{R}{L+1},\qquad \mathsf{I}_Q(A)=\max\Big\{1,\frac{R-1}{L}+O(|A|^{-1})\Big\}.$$

Thus $\mathsf{D}^{\mathrm{av}}_Q(A)\mathsf{I}^{\mathrm{av}}_Q(A)=R$ exactly. Taking $L\asymp R^{1/2}$ makes both average factors, and both maximum factors, of order $R^{1/2}$. Hence one cannot generally force either named factor to absorb more than half of the logarithmic doubling exponent.

Raw one-tape time alone cannot provide such a constraint uniformly across predicates. For the easy predicate $A_0=\{x\in\{0,1\}^N : x_1=0\}$, the map

$$Q_0(x)=\begin{cases}0^{N+1}, & x_1=x_2=0,\\ 0x, & x_1=0,\ x_2=1,\\ 1x, & x_1=1\end{cases}$$

is computable by a linear-time scan and exactly preserves the predicate "first output bit is zero.'' Its source-YES fibers consist of one fiber of size 2^{N-2} and 2^{N-2} singletons, so $\overline{F}<2$ while $\mathsf{I}_{Q_0}(A_0)>2^{N-3}$. Thus a runtime-to-geometry theorem must use MCSP-specific semantics. For the predicate map $Q_A=1_A$, computing Q_A is time-equivalent, up to output materialization, to deciding A; applying such a theorem to Q_A would itself constitute the desired MCSP time lower bound.

A non-dummy tensor test

Theorem 1 is one-sided and general. A natural non-dummy family shows that equality is forced when both sides of the promise are preserved. Fix an integer $t\ge 2$ and a Boolean function $\phi:\{0,1\}^t\to\{0,1\}$ with $\phi(0^t)=0$ and $\phi(1^t)=1$, and define

$$E_{t,\phi}(f)(x^{(1)},\dots,x^{(t)})=\phi\big(f(x^{(1)}),\dots,f(x^{(t)})\big). \tag{3m}$$

This encoding has $n'=tn$ and is exactly decodable on the diagonal:

$$E_{t,\phi}(f)(x,\dots,x)=f(x). \tag{3n}$$

Consequently,

$$\mathrm{CC}(f)\le\mathrm{CC}(E_{t,\phi}(f))\le t\,\mathrm{CC}(f)+O_\phi(1). \tag{3o}$$

The left inequality restricts all t input blocks to the same variables, and the right one uses t copies of a circuit for f followed by a fixed circuit for ϕ.

For completeness, circuit complexities are exponentially dense: whenever $0<\alpha<\beta<1$, all sufficiently large n admit an n-variable function f with

$$2^{\alpha n}<\mathrm{CC}(f)<2^{\beta n}. \tag{3p}$$

To see this, choose $\alpha<\gamma<\beta$ and consider functions on $\lfloor\gamma n\rfloor$ variables. Circuit counting puts one above $2^{\alpha n}$, while the DNF upper bound puts every such function below $2^{\beta n}$.

Now require, for every f, both

$$\mathrm{CC}(f)\le 2^{\mu_H n}\Longrightarrow\mathrm{CC}(E_{t,\phi}(f))\le 2^{t\mu_1 n} \tag{3q}$$

and the reverse NO implication with both inequalities negated. If $\mu_H>t\mu_1$, (3p) supplies a source YES instance whose encoded complexity exceeds the target threshold by (3o). If $\mu_H<t\mu_1$, it supplies a source NO instance of complexity below $2^{\gamma n}$, for some $\mu_H<\gamma<\min\{t\mu_1,1\}$; the upper bound in (3o) makes its encoding a target YES instance. Therefore

$$\boxed{\mu_H=t\mu_1.} \tag{3r}$$

Thus direct products and tensors do not improve on the threshold-per-log-length ratio preserved by dummy padding.

Exact threshold alignment by dummy variables

To test the fixed endpoint (2) by dummy variables, specialize the free target exponent in this section and the calculation of the next section to $\mu_1 = \mu_*$. Write μ_1 in the formulas only to keep the general source/target threshold notation visible.

Let

$$k = \min\{j \in \mathbb{N} : j\mu_1 > \tfrac{1}{2}\}, \qquad \mu_H = k\mu_1. \tag{4}$$

Then

$$\tfrac{1}{2} < \mu_H \le \tfrac{1}{2} + \mu_1 < \tfrac{3}{4}. \tag{5}$$

For $f : \{0,1\}^n \to \{0,1\}$, define

$$E_k(f)(x,z) = f(x), \qquad x \in \{0,1\}^n, \quad z \in \{0,1\}^{(k-1)n}. \tag{6}$$

In the bounded-fan-in, unbounded-fan-out De Morgan basis,

$$\mathrm{CC}(E_k(f)) = \mathrm{CC}(f). \tag{7}$$

The forward inequality ignores the dummy variables. Conversely, replacing every dummy-input leaf in a circuit for $E_k(f)$ by an x_1-leaf yields a circuit for f without increasing size. Since $N' = 2^{kn} = N^k$,

$$f \in L_{\mu_H,n} \quad \Longleftrightarrow \quad E_k(f) \in L_{\mu_1,kn}. \tag{8}$$

Thus the threshold alignment itself is exact; it is not the missing interface.

The unavoidable exponent loss

The oracle-strengthened CHMY theorem [2, Theorem 3] implies that for every $1/2 < \lambda < \mu_H$,

$$L_{\mu_H} \notin \mathrm{BPTIME}_1[N^{2(\lambda - o(1))}]. \tag{9}$$

Suppose, optimistically, that a low-threshold decider running in time $(N')^a$ could be composed with (6) without any additional exponent loss. Then the induced high-threshold decider would run in

$$(N')^a = N^{ka}. \tag{10}$$

Combining (9) and (10) rules out only

$$a < \frac{2\lambda}{k} - o(1). \tag{11}$$

Letting $\lambda \uparrow \mu_H = k\mu_1$, the best possible endpoint is

$$a < 2\mu_1 - o(1) < \tfrac{1}{2}. \tag{12}$$

This is strictly below 1.01, the scale required in (2). Equivalently, to rule out $(N')^{1.01}$ by (9), one would need

$$1.01k < 2\lambda < 2k\mu_1, \qquad \text{hence} \qquad \mu_1 > 0.505, \tag{13}$$

contradicting (3).

An ordinary materialized reduction is weaker still: it must write all $N' = N^k$ padded bits. Since $\mu_1 < 1/4$, we have $k \ge 3$, and the output-generation cost already lies above the time scale that (9) could exclude.

Conditional composition boundaries

The idealized calculation above is deliberately generous. In the canonical table of $E_k(f)$, each source truth-table bit is repeated

$$R = N^{k-1} \tag{14}$$

times. An online one-tape generator must know the final input length before emitting the first repeated block. Scanning the whole source input and then simulating a target machine requires storing the source table and a one-tape work simulation. The MMW and CHMY simulation lemmas do not provide an $N'^{o(1)}$-overhead guarantee for this particular composition.

Therefore (12) is an optimistic upper bound for the padding route: even if this additional composition interface were supplied for free, the exponent would remain too small. Corollary 2 shows that the same conclusion holds for every YES-preserving map satisfying the small-fiber condition (3h), before this composition issue arises.

Proposition 8 (the materialization boundary for arbitrary quotients)

Fix constants $a, b, \kappa > 0$ and $1/2 < \mu_H < 1$, and let Q_n be a deterministic, pointwise exact threshold reduction with $N' = N^{\kappa+o(1)}$. Assume explicitly that a Q_n-materializer taking time $N^{b+o(1)}$ and a target decider taking time $(N')^{a+o(1)}$ compose into a CHMY-model source machine with total time

$$N^{o(1)}\left(N^{b+o(1)} + (N')^{a+o(1)}\right).$$

Here both truth tables use canonical order; work-tape initialization, target-input delivery, and head relocation are included in the handoff. If the target machine is randomized, its error must be preserved; if the oracle-strengthened source lower bound is invoked, every composed oracle query must still have length $N^{o(1)}$. Then the composed source computation has exponent

$$\beta = \max\{b, \kappa a\}. \tag{15}$$

It enters some CHMY lower-bound window with $1/2 < \lambda < \mu_H$ if and only if

$$\boxed{\beta < 2\mu_H.} \tag{16}$$

In a separate fresh-output transducer model with an initially blank sequential output, at most $O(1)$ output bits written per step, and all canonical N' bits explicitly written, one has $b \geq \kappa$. Hence output length alone closes every CHMY window exactly when $\kappa \geq 2\mu_H$. If materialization is optimal, $b = \kappa$, and $a \geq 1$, the surviving window is instead

$$\boxed{\kappa a < 2\mu_H.} \tag{17}$$

Indeed, the CHMY bound (9) contradicts the composed upper bound precisely when $\beta < 2\lambda$ for some $1/2 < \lambda < \mu_H$, which is equivalent to (16). The remaining claims follow from the fresh-output lower bound N' and (15). □

This is a conditional exponent boundary, not a composition lemma supplied by MMW or CHMY. MMW's one-pass algorithm is a RAM-model streaming algorithm on the original canonical input. CHMY's DTIME_1 / BPTIME_1 model has a one-way read-only input tape and one two-way read/write work tape, with no independent materialized-output interface; its streaming simulation lemma acts on the same given input stream.

Proposition 9 (an exact switch quotient forces the missing branch bit; an illustrative example)

Fix an integer $k \geq 1$, put $N' = N^k$, $S = \lfloor N^{\mu_H} \rfloor$, and $T = \lfloor N^{k\mu_1} \rfloor$, and suppose $1/2 < \mu_H < 1$ and $\mu_H > k\mu_1$. Let E_k be the dummy lift from (6), and define

$$Q_n^{\mathrm{sw}}(f) = \begin{cases} 0^{N'}, & \mathrm{CC}(f) \leq S, \\ E_k(f), & \mathrm{CC}(f) > S. \end{cases}$$

Then, for all sufficiently large n,

$$f \in L_{\mu_H,n} \quad \Longleftrightarrow \quad Q_n^{\mathrm{sw}}(f) \in L_{\mu_1,kn}.$$

Moreover, if a deterministic CHMY-compatible fresh-output transducer materializes all N' output bits in time $N^{b+o(1)}$, then

$$\boxed{\max\{b, k\} \geq 2\mu_H.}$$

Indeed, $0^{N'}$ has constant circuit complexity. On the NO branch, $\mathrm{CC}(E_k(f)) = \mathrm{CC}(f) > S > T$, so the output cannot be $0^{N'}$. Running the transducer and accepting exactly when every emitted bit is zero therefore decides $L_{\mu_H,n}$ in one-tape time $N^{\max\{b,k\}+o(1)}$. If the displayed inequality failed, choose $1/2 < \lambda < \mu_H$ with $\max\{b,k\} < 2\lambda$ and contradict the CHMY lower bound (9). This is a lower bound for the deliberately branch-forcing quotient Q_n^{sw}, not for arbitrary predicate-specific quotients; it shows why an MMW search output alone cannot supply the missing exact materialization interface.

Limitations and scope

For constant log-length expansion $n' = \kappa n + O(1)$, the counting inequality (3d) closes every truth-table map whose easy-function fibers have logarithmic size $o(n2^{\mu_H n})$, including dummy padding, exactly decodable encodings, tensor lifts and quotients by $2^{\mathrm{poly}(n)}$-size symmetry families. Proposition 6 closes every exact coset-fiber quotient. The surviving exact possibility uses non-coset, predicate-specific fibers, which Proposition 7 forces to have exponential XOR-defect or fiber inflation; the decision-bit map shows such maps exist combinatorially, so ruling them out requires bounding the one-tape cost of computing them without deciding source MCSP. The exact stabilizer of $\mathcal{L}_S$ is not determined. For gap versions, the case $S_1 \geq 2S_0$ and kernels containing source-threshold functions are open. Propositions 8–9 are conditional on, or specific to, explicit composition models. Nothing here proves $P \neq NP$ or a new lower bound.

Reproducibility

All results are proved in the text. The supplement contains `checks.py` (Python 3 standard library), which performs exhaustive finite checks on tiny parameters: exact circuit complexities for $n \leq 2$ and for all functions of size ≤ 5 on $n = 3$; point-indicator and point-flip costs; Harper's theorem on the 4-cube (all subsets) and its iterated form for the sets $\mathcal{L}_t$ in the 8-cube; the downset edge identity in (39); the coset-quotient doubling bound (52); the identities and inequalities (58), (59), (55) for random maps; the predicate-map and Q_0 examples; and the tensor and padding identities. Run `python3 checks.py` (expected output: `1048 checks passed`). These are numerical sanity checks, not proofs of the asymptotic statements.

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